

1 Preliminary notions

1.1 Probability theory

The foundation of modern probability theory is based on measure theory.

Definition 1.1 (σ -algebra). A σ -algebra $\mathcal{F}$ on a set Ω is a collection of subsets of Ω that

- $\Omega \in \mathcal{F}$
- if $A \in \mathcal{F}$ then $\Omega \setminus A \in \mathcal{F}$
- if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Definition 1.2 (Measure). Let Ω be a set and $\mathcal{F}$ a σ -algebra on Ω . A function $\mu : \mathcal{F} \rightarrow [0, \infty]$ is **measure** if satisfies

1. $\mu(\emptyset) = 0$
2. $\mu(A) \geq 0$ for all $A \in \mathcal{F}$
3. if $\{A_i\}_{i=1}^{\infty} \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$ for all $i \neq j$ then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

If μ is a measure on $\mathcal{F}$ such that $\mu(\Omega) = 1$, then μ is called a **probability measure**.

Definition 1.3 (Probability space). A probability space is a triple $(\Omega, \mathcal{F}, P)$ where

- Ω is the set of all possible outcomes
- $\mathcal{F}$ is a σ -algebra on Ω
- P is a probability measure.

For a more details in probability theory and stochastic processes, see [8] or [9].

1.2 Stochastic processes

Definition 1.4 (Stochastic processes). A stochastic process is an indexed family of random variables

$$\{X(t, \omega)\}_{t \in T},$$

on a probability space $(\Omega, \mathcal{F}, P)$ assuming values in $\mathbb{R}^n$.

If the set T is countable then the stochastic process is said to be *discrete*, if T is some interval of $\mathbb{R}$ then it is *continuous*.

An intuitive way to understand stochastic processes is to consider what is $X(t, \omega)$ when t or ω is fixed. If we fix $t \in T$ then the map

$$\omega \rightarrow X_t(\omega),$$

for $\omega \in \Omega$ is a random variable. If $\omega \in \Omega$ is fixed then the map

$$t \rightarrow X_\omega(t),$$

for $t \in T$ is a deterministic function, and is called *path* of $X(t, \omega)$.

Definition 1.5 (Filtration). A **filtration** is a family of σ -algebras $\{\mathcal{F}_t\}_{t \in T}$ indexed by T , such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$, where $s, t \in T$.

The idea is that a filtration $\mathcal{F}_t$ represents the information available from the past of an stochastic process up to time t . Thus we define a **filtered probability space** as a probability space with a filtration $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, P)$.

Definition 1.6 (Adapted process). Let X_t be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, P)$. The process is **adapted** to the filtration $\{\mathcal{F}_t\}_{t \in T}$ if, for every $t \in T$, the random variable X_t is $\mathcal{F}_t$ -measurable.

The idea of an adapted process is key when modeling using stochastic processes, as it ensures that the process at time t depends only on the information available up to that time, but not on future information.

1.2.1 Martingales

A martingale is a fundamental concept in the theory of stochastic processes, formalizes the idea of a "fair game", where given all the information up to now, the best prediction for the future is simply the present value.

Definition 1.7 (Martingale). A stochastic process X_t where $t \in T$ is called a **martingale** with respect to a filtration $\{\mathcal{F}_t\}_{t \in T}$ if it satisfies the following conditions:

1. $E[|X_t|] < \infty$, for each $t \in T$
2. X_t is measurable with respect to $\mathcal{F}_t$ for each $t \in T$
3. If X_t is discrete

$$E[X_{t+1}|\mathcal{F}_t] = X_t \quad \forall t \in \mathbb{N},$$

if X_t is continuous

$$E[X_t|\mathcal{F}_s] = X_s \quad \forall s \leq t.$$

1.2.2 Brownian Motion

Brownian motion (also called Wiener process) is the fundamental stochastic process underlying continuous-time financial modeling. Formally, a Brownian motion $\{W_t\}_{(t \geq 0)}$ is defined by the following properties:

1. $W_0 = 0$
2. The map $t \rightarrow W_t(\omega)$ is almost surely continuous
3. W_t has independent increments, i.e. for $t_1 < t_2 < t_3 < t_4$, $W_{t_2} - W_{t_1}$ and $W_{t_4} - W_{t_3}$ are independent random variables
4. W_t has gaussian increments, i.e. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s \leq t$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2 .

The Brownian motion has some key features that will need afterwards, for example it can be shown that the paths $t \rightarrow W_t(\omega)$ although are almost surely continuous, they are nowhere differentiable.

Also it can be shown that the Wiener process is an almost surely continuous martingale with quadratic variation $[W, W]_t = t$. These two properties together with $W_0 = 0$ form an equivalent definition of the Wiener process, known as Lévy's characterization.

Proofs of these properties, the equivalence between the two definitions, and the existence of Brownian motion can be found in [10] and [11, Ch. 2].

2 Stochastic calculus

To model the dynamics of a financial asset's value, an ordinary differential equation of the form

$$\frac{dS_t}{dt} = f(t, S_t)$$

is insufficient: it cannot capture the random fluctuations inherent in financial data. Stochastic differential equations (SDEs) add a noise term to handle this,

$$dS_t = f(t, S_t) dt + \sigma(t, S_t) dW_t, \quad (1)$$

where W_t is a Brownian motion. In integral form this reads

$$S_t = S_0 + \int_0^t f(s, S_s) ds + \int_0^t \sigma(s, S_s) dW_s, \quad (2)$$

thus a key step is to give rigorous meaning to the stochastic integral $\int \sigma dW_s$.

The most well known process that has a well defined stochastic integral is the Wiener process or Brownian motion, this stochastic integral is known as the Ito integral and will allow us to have the machinery to study SDEs and thus model the dynamics of financial assets.

2.1 Ito's integral

Definition 2.1. Let $\mathcal{V} = \mathcal{V}(S, T)$ the class of functions

$$f(t, \omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R},$$

such that

- (t, ω) is $\mathcal{B} \times \mathcal{F}$ -measurable, where $\mathcal{B}$ is the Borel σ -algebra on $[0, \infty)$ and $\mathcal{F}$ is the σ -algebra on Ω ,
- $f(t, \omega)$ is $\mathcal{F}_t$ adapted.
- $\mathbb{E} \left[\int_S^T f(t, \omega)^2 dt \right] < \infty$.

We will start by defining the Itô integral for *elementary functions*, then this can be extended to functions in $\mathcal{V}$.

Definition 2.2. A function ϕ is called *elementary* if it has the form

$$\phi(t, \omega) = \sum_j e_j(\omega) \mathbf{1}_{[t_j, t_{j+1})}(t).$$

For elementary functions we define the Ito integral as

$$\int_S^T \phi(t, \omega) dB_t(\omega) = \sum_j e_j(\omega) [B_{t_{j+1}} - B_{t_j}](\omega).$$

Lemma 2.1 (Ito isometry). Let $\phi(t, \omega)$ be a bounded elementary function, then

$$\mathbb{E} \left[\left(\int_S^T \phi(t, \omega) dB_t(\omega) \right)^2 \right] = \mathbb{E} \left[\int_S^T \phi(t, \omega)^2 dt \right].$$

Using the Ito isometry and the definition of the Ito integral for elementary functions, the Ito integral can be extended to the class of functions $\mathcal{V}$, using that the functions in $\mathcal{V}$ can be expressed as the limit of sequences of simpler functions. It is worth mentioning that the Itô isometry still holds for any function in $\mathcal{V}$.

The result of the integral is a stochastic process, which is a continuous martingale. This is similar to regular integration, where the result of integrating a function is a new function, in this case the result of integrating a process "over" a stochastic process is a new stochastic process, plus the resulting process has the nice property of being a martingale.

The class $\mathcal{V}$ is rather limited and does not even contain all continuous functions, see [12]. Having to restrict to this class would be rather inconvenient when modeling using SDEs, luckily the Itô integral can be extended to a broader class by relaxing the last condition of Definition 2.1

$$P\left(\int_S^T f(t, \omega)^2 dt < \infty\right) = 1,$$

will call this class of functions $\mathcal{W}$. Notice that the class $\mathcal{W}$ does contain all continuous functions.

2.2 Ito's formula

Having defined the Ito integral, would be interesting to have a way to easily compute stochastic integrals without having to compute the limit of elementary functions, that is the purpose of the Ito's lemma or Ito's formula. The Ito's formula is a stochastic version of the chain rule for derivatives.

Definition 2.3 (Ito process). Let B_t be a Brownian motion on $(\Omega, \mathcal{F}, P)$. An Ito process is a stochastic process X_t on $(\Omega, \mathcal{F}, P)$ of the form

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s, \quad (3)$$

where μ and σ are adapted, measurable processes, that satisfy the integrability conditions

$$P\left(\int_0^T |\mu(t, \omega)| dt < \infty\right) = 1,$$

and

$$P\left(\int_0^T \sigma(t, \omega)^2 dt < \infty\right) = 1.$$

The equation (3) can be also written in differential form as

$$dX_t = \mu dt + \sigma dB_t.$$

Theorem 2.2 (1-dimensional Itô's formula). Let X_t be an Itô process given by

$$dX_t = \mu dt + \sigma dB_t.$$

Let $g(t, x) \in C^2([0, \infty) \times \mathbb{R})$, then

$$Y_t = g(t, X_t),$$

is an Ito process given by

$$dY_t = \frac{\partial g}{\partial t}(t, X_t)dt + \frac{\partial g}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2}(t, X_t)(dX_t)^2, \quad (4)$$

where $(dX_t)^2$ is computed according to the rules

$$(dB_t)^2 = dt, \quad (dt)^2 = 0, \quad dB_t \cdot dt = 0.$$

For the proof of the Ito's formula we refer to [11]. The formula can be extended to multiple dimensions.

Definition 2.4 (n -dimensional Ito process). Let $B_t = (B_1(t), \dots, B_m(t))$ be a m -dimensional Brownian motion, let μ_i and σ_{ij} be adapted, measurable processes such that

$$P\left(\int_0^T |\mu_i(t, \omega)| dt < \infty\right) = 1, \quad P\left(\int_0^T \sigma_{ij}(t, \omega)^2 dt < \infty\right) = 1,$$

then we define the n -dimensional Ito process $X_t = (X_1(t), \dots, X_n(t))$ as

$$\begin{cases} dX_1(t) = \mu_1 dt + \sigma_{11} dB_1(t) + \dots + \sigma_{1m} dB_m(t), \\ \vdots \\ dX_n(t) = \mu_n dt + \sigma_{n1} dB_1(t) + \dots + \sigma_{nm} dB_m(t), \end{cases}$$

or in matrix notation

$$dX_t = \mu dt + \sigma dB_t,$$

where $\mu = (\mu_1, \dots, \mu_n)$ and $\sigma = (\sigma_{ij})$ are an n -dimensional vector and an $n \times m$ matrix respectively.

Theorem 2.3 (General Itô's formula). Let X_t be an n -dimensional Itô process given by

$$dX_t = \mu dt + \sigma dB_t.$$

Let $g(t, x) = (g_1(t, x), \dots, g_p(t, x)) \in C^2([0, \infty) \times \mathbb{R}^n)$, then

$$Y_t = g(t, X_t),$$

is a p -dimensional Ito process whose components are given by

$$dY_k(t) = \frac{\partial g_k}{\partial t}(t, X_t)dt + \sum_i \frac{\partial g_k}{\partial x_i}(t, X_t)dX_i(t) + \frac{1}{2} \sum_{ij} \frac{\partial^2 g_k}{\partial x_i \partial x_j}(t, X_t)dX_i(t)dX_j(t), \quad (5)$$

where $dX_i(t)dX_j(t)$ is computed according to the rules

$$(dB_i(t))^2 = dt, \quad dB_i(t)dB_j(t) = 0 \quad (i \neq j).$$

2.3 Stochastic differential equations

A **stochastic differential equation** (SDE) is an equation of the form

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 = x_0, \quad (6)$$

where $b : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the *drift* and $\sigma : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is the *diffusion* coefficient. A solution is an Itô process X_t satisfying (6) almost surely.

The following theorem, whose proof may be found in [11, Thm. 5.2.1], gives sufficient conditions for a solution to exist and be unique.

Theorem 2.4 (Existence and uniqueness). Let $T > 0$. Suppose b and σ satisfy, for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,

1. (Lipschitz) $|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq C|x - y|$,
2. (Linear growth) $|b(t, x)|^2 + |\sigma(t, x)|^2 \leq C^2(1 + |x|^2)$,

for some constant $C > 0$, and let $x_0 \in \mathbb{R}^n$. Then the SDE (6) has a unique (up to indistinguishability) continuous adapted solution $\{X_t\}_{0 \leq t \leq T}$.

We close with two fundamental examples, both following [11, Ch. 5].

2.3.1 Example: Geometric Brownian motion

Let $\mu, \sigma \in \mathbb{R}$ with $\sigma \neq 0$ and $S_0 > 0$. Consider the SDE

$$dS_t = \mu S_t dt + \sigma S_t dB_t. \quad (7)$$

This is called **geometric Brownian motion** and is the model used in Section 3.2 for the dynamics of a stock price.

Applying Theorem 2.2 to $g(t, x) = \ln x$, with $g_t = 0$, $g_x = x^{-1}$, $g_{xx} = -x^{-2}$,

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 = \mu dt + \sigma dB_t - \frac{\sigma^2}{2} dt = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dB_t.$$

Integrating both sides from 0 to t gives $\ln(S_t/S_0) = (\mu - \sigma^2/2)t + \sigma B_t$, so the unique solution is

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B_t \right]. \quad (8)$$

2.3.2 Example: Ornstein-Uhlenbeck process

Let $\theta, \sigma > 0$ and $\mu \in \mathbb{R}$. Consider the SDE

$$dX_t = \theta(\mu - X_t) dt + \sigma dB_t, \quad X_0 = x_0. \quad (9)$$

This is the **Ornstein-Uhlenbeck (OU) process**, a mean-reverting model: the drift $\theta(\mu - X_t)$ pulls the process toward the long-run mean μ at rate θ . It appears, for instance, in the Vasicek interest rate model.

To solve, let $Y_t = (X_t - \mu)e^{\theta t}$. Applying Theorem 2.2 with $g(t, x) = (x - \mu)e^{\theta t}$,

$$dY_t = \theta(X_t - \mu)e^{\theta t} dt + e^{\theta t} dX_t = e^{\theta t} [\theta(\mu - X_t) + \theta(X_t - \mu)] dt + \sigma e^{\theta t} dB_t = \sigma e^{\theta t} dB_t.$$

Integrating and reverting the substitution,

$$X_t = \mu + (x_0 - \mu) e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dB_s. \quad (10)$$

The OU process is Gaussian with $\mathbb{E}[X_t] = \mu + (x_0 - \mu)e^{-\theta t} \rightarrow \mu$ as $t \rightarrow \infty$.

3 Black-Scholes

Black-Scholes is a mathematical model for pricing options. Although many other models have been developed since, the Black-Scholes model remains as the most important model in option pricing, as in many cases options in markets are quoted using the Black-Scholes formula.

A derivative is a financial instrument whose payments are derived from the price of an other asset, called the underlying asset. Derivatives are widely used in finance, whether to hedge financial risk, this is to protect against adverse price movements, or to bet on specific views, especially when it's difficult to take a position in the underlying asset. So being able to price them and justify the price is key for the market to function.

3.1 Replication and arbitrage

One of the most important concepts in modern finance is the so called *law of one price*. The idea is that if two financial assets have exactly the same payoffs in all possible scenarios, then they should have the same price.

This is linked to the concept of *arbitrage*, informally we say that arbitrage is the opportunity to make a riskless profit. For example, if two assets that have exactly the same payoffs in all possible scenarios, but don't follow the law of one price, one can buy the cheaper one and sell the more expensive one, gaining an instant profit and compensating the payoffs of the sold asset using the payoffs of the bought one, which match exactly.

We will work under the assumption that we are operating in a market with no arbitrage, this assumption is technically known as *The First Fundamental Theorem of Asset Pricing*, and a key part of the *Efficient Market Hypothesis*. Under this assumption, we can price any derivative by creating a replicating portfolio, this is a set of assets that has exactly the same payoffs as the derivative in all possible scenarios.

There are two main ways of creating a replicating portfolio:

- **Static replication** refers to an replicating portfolio that is created at the beginning and held as is until maturity.
- **Dynamic replication** refers to a replicating portfolio that requires a continuous change of its composition over time.

For a further discussion on replication, and its applications in finance, we recommend the book [4].

We will show two examples of static replication from [12] that will be useful to understand the Black-Scholes model.

3.1.1 Example: Forward contract

Assume that there is an agent that borrows and lends money at a continuously compounded rate r , known as the **risk-free rate**. This means that if we borrow 1\$ at time $t = 0$, we will have to pay e^{rT} \$ at time $t = T$.

A forward contract gives the holder the obligation to buy an asset at a fixed price K at a fixed date T .

To price this contract imagine we are at time t and the current price of an asset is S , we want to buy a forward contract with maturity T and fixed price K .

We have two options, we can buy the contract at time t for a price of F and at time T pay K to get the asset. Alternatively we can buy the asset for S and borrow $e^{-r(T-t)}K$

at time t , so at time T we will pay $K = e^{r(T-t)}e^{-r(T-t)}K$ while having the asset. Both options have the same cashflow at time T , pay K and own the asset, so either we have

$$F = S - e^{-r(T-t)}K,$$

or there is an arbitrage opportunity.

We have found the price of the forward contract by replicating its cashflows, using the underlying asset and the risk-free rate, plus arguing that there should not be an arbitrage opportunity.

3.1.2 Example: Put-Call parity

Definition 3.1 (European call option). An **European call option** is a financial contract that gives the holder the right, but not the obligation, to **buy** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European call option at maturity T is given by:

$$C_T = \max(S_T - K, 0), \quad (11)$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

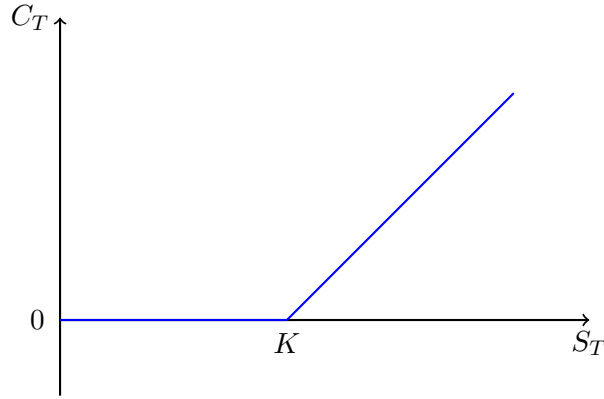


Figure 1: Payoff of a European call option

Definition 3.2 (European put option). An **European put option** is a financial contract that gives the holder the right, but not the obligation, to **sell** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European put option at maturity T is given by:

$$P_T = \max(K - S_T, 0), \quad (12)$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

Now consider a portfolio where we buy a call option and sell a put option with same strike K and maturity T , the payoff of such portfolio at time T will be exactly $S_T - K$. So we are essentially receiving the asset and paying K at time T , as we have seen this is the payoff of a forward contract, which means that at time t the price of our portfolio must be equal to the price of a forward of the asset,

$$C_t - P_t = F_t = S_t - e^{-r(T-t)}K, \quad (13)$$

this formula is called *put-call parity* and illustrates the relationship between the price of a call and a put option.

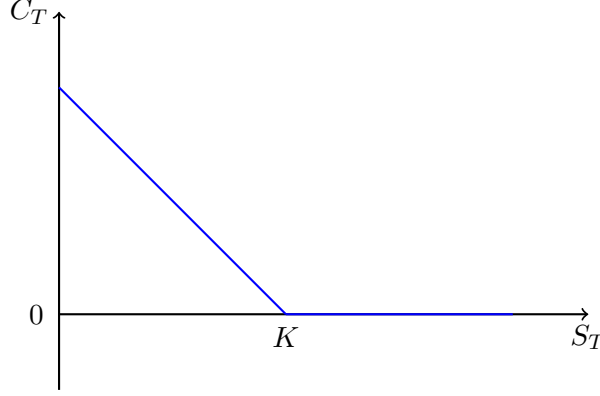


Figure 2: Payoff of a European put option

3.2 The Black-Scholes model

The Black-Scholes model is a mathematical model for pricing options, that is based on replicating the payoffs of the option dynamically using the underlying asset and a risk-free rate bond.

Like before we assume that the market is arbitrage-free, and that there is a constant risk-free rate r . As the model is based on dynamic replication, we also need to assume that we can rebalance our replicating portfolio continuously, i.e. we can buy and sell any amount of the underlying asset and the risk-free rate bond instantaneously at any time.

Let S_t be the price of the underlying asset at time t and β_t be the price of the risk-free rate bond at time t . The dynamics of this two assets are given by the following stochastic differential equations (SDEs):

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad d\beta_t = r\beta_t dt.$$

To construct the replicating portfolio. Let a_t be the amount of the underlying asset in our replicating portfolio and b_t be the amount of the risk-free rate bond in our replicating portfolio. Then the value of the portfolio at any time t is given by:

$$V_t = a_t S_t + b_t \beta_t.$$

We need this portfolio to have the same payoff as the option at time T , so

$$V_T = \max(S_T - K, 0).$$

As the portfolio is dynamically replicating the option, the cashflow of the portfolio must match the ones of the option, these are the payoff as seen before, and the initial payment, so no cash must come in or out of the portfolio at any time between t and T . This means that the portfolio must be *self-financing*, so the rebalanced must come from selling the underlying asset and buying the bond, or vice versa. This is expressed mathematically as

$$dV_t = a_t dS_t + b_t d\beta_t, \tag{14}$$

which can be extended to

$$dV_t = (a_t \mu S_t + b_t r \beta_t) dt + a_t \sigma S_t dW_t.$$

Now as the value of the replicating portfolio V_t should be equal to the value of the option at any time t , we assume that this can be expressed by a function $f(t, S_t)$, depending only on the time t and the price of the underlying asset S_t . Thus applying Ito's formula we have that

$$dV_t = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} \mu S_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right) dt + \frac{\partial f}{\partial S} \sigma S_t dW_t,$$

matching the two expressions for dV_t we have that

$$a_t = \frac{\partial f}{\partial S},$$

and

$$b_t = \frac{1}{r\beta_t} \left(\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right)$$

as V_t is the value of the replicating portfolio and the option at time t , we have that

$$f(t, S_t) = V_t = a_t S_t + b_t \beta_t,$$

using the values of a_t and b_t from above,

$$f(t, S_t) = \frac{\partial f}{\partial S} S_t + \frac{1}{r} \left(\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right),$$

that can be rearranged to

$$\frac{\partial f}{\partial t} = r f(t, S_t) - r \frac{\partial f}{\partial S} S_t - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2, \quad (15)$$

with terminal condition

$$V_T = f(T, S_T) = \max(S_T - K, 0),$$

which is the **Black-Scholes PDE**.

3.3 Pricing - Solving the Black-Scholes PDE

There is an analytical solution to the Black-Scholes PDE. Our derivation follows the approach of [12, Ch. 11.4], adapted to log-moneyness coordinates $x = \ln(S/K)$ instead of the log-price $\ln S$ used there; the two differ only by the constant shift $\ln K$ and the subsequent reduction to the heat equation is structurally identical.

Start by change of variables. Let $\tau = T - t$ be the time to maturity, and $x = \ln\left(\frac{S}{K}\right)$ be the log-moneyness. We define a new function $u(\tau, x) = f(t, S)$ so the Black-Scholes PDE is transformed to,

$$\frac{\partial u}{\partial \tau} = \frac{1}{2} \sigma^2 \frac{\partial^2 u}{\partial x^2} + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\partial u}{\partial x} - r u(\tau, x).$$

Now we would like to remove the drift term (the derivative with respect to x) and the zero order term, so we have a standard heat equation. To do it we do another transformation,

$$u(x, \tau) = e^{\alpha x + \beta \tau} v(x, \tau),$$

where α and β are the coefficients that lead to the heat equation, using,

$$\begin{aligned} u_\tau &= e^{\alpha x + \beta \tau}(\beta v + v_\tau), \\ u_x &= e^{\alpha x + \beta \tau}(\alpha v + v_x), \\ u_{xx} &= e^{\alpha x + \beta \tau}(\alpha^2 v + 2\alpha v_x + v_{xx}), \end{aligned}$$

we get,

$$v_\tau = \frac{1}{2}\sigma^2 v_{xx} + v_x \left(\sigma^2 \alpha + r - \frac{1}{2}\sigma^2 \right) + v \left(\frac{1}{2}\sigma^2(\alpha^2 - \alpha) + r\alpha - r - \beta \right),$$

thus,

$$\alpha = \frac{1}{2} - \frac{r}{\sigma^2} = -\frac{1}{\sigma^2} \left(r - \frac{1}{2}\sigma^2 \right),$$

and

$$\begin{aligned} \beta &= \alpha \left(r + \frac{1}{2}\sigma^2(\alpha - 1) \right) - r \\ &= \alpha \left(\frac{1}{2}r - \frac{1}{4}\sigma^2 \right) - r \\ &= -\frac{1}{2\sigma^2} \left(r - \frac{1}{2}\sigma^2 \right)^2 - r \\ &= -\frac{1}{2\sigma^2} \left(r + \frac{1}{2}\sigma^2 \right)^2. \end{aligned}$$

So we get the standard heat equation,

$$\frac{\partial v}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 v}{\partial x^2},$$

with initial conditions for v ,

$$v(x, 0) = e^{-\alpha x} u(x, 0) = K e^{-\alpha x} \max(e^x - 1, 0).$$

Now we can solve the equation using the heat operator,

$$v(x, \tau) = \int_{-\infty}^{\infty} v(y, 0) k(\sigma^2; \tau, x - y) dy,$$

where the function k is the *heat kernel*, i.e. the fundamental solution of the heat equation $v_\tau = \lambda v_{xx}$, given by

$$k(\lambda; t, x) := \frac{1}{\sqrt{4\lambda\pi t}} \exp\left(-\frac{x^2}{4\lambda t}\right), \quad (16)$$

see [12, Ch. 11.4]. Using the initial condition we get,

$$v(x, \tau) = \int_0^\infty K e^{-\alpha y} (e^y - 1) k\left(\frac{\sigma^2}{2}; \tau, x - y\right) dy,$$

this leaves us with two integrals

$$\begin{aligned} I_1 &= K \int_0^\infty e^{(1-\alpha)y} k\left(\frac{\sigma^2}{2}; \tau, x - y\right) dy, \\ I_2 &= -K \int_0^\infty e^{-\alpha y} k\left(\frac{\sigma^2}{2}; \tau, x - y\right) dy, \end{aligned}$$

if we make a change of variable, $s = x - y$, then

$$\begin{aligned} I_1 &= K e^{(1-\alpha)x} \int_{-\infty}^x e^{-(1-\alpha)s} k\left(\frac{\sigma^2}{2}; \tau, s\right) ds, \\ I_2 &= -K e^{-\alpha x} \int_{-\infty}^x e^{\alpha s} k\left(\frac{\sigma^2}{2}; \tau, s\right) ds. \end{aligned}$$

To evaluate both integrals we use the following identity, whose proof is a standard completion of the square.

Lemma 3.1. Let Φ denote the standard Gaussian cumulative distribution function,

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

Then for any $a, b \in \mathbb{R}$ and $\sigma, \tau > 0$,

$$\int_{-\infty}^a k\left(\frac{\sigma^2}{2}; \tau, s\right) e^{-bs} ds = e^{\frac{b^2 \sigma^2 \tau}{2}} \Phi\left(\frac{a}{\sigma \sqrt{\tau}} + b \sigma \sqrt{\tau}\right). \quad (17)$$

Proof. Completing the square in the exponent of the integrand,

$$-\frac{s^2}{2\sigma^2\tau} - bs = -\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau} + \frac{b^2\sigma^2\tau}{2},$$

so pulling out the constant factor and applying the change of variable $u = (s + b\sigma^2\tau)/(\sigma\sqrt{\tau})$,

$$\begin{aligned} \int_{-\infty}^a k\left(\frac{\sigma^2}{2}; \tau, s\right) e^{-bs} ds &= e^{\frac{b^2 \sigma^2 \tau}{2}} \int_{-\infty}^a \frac{1}{\sqrt{2\pi}\sigma^2\tau} \exp\left(-\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau}\right) ds \\ &= e^{\frac{b^2 \sigma^2 \tau}{2}} \int_{-\infty}^{\frac{a + b\sigma^2\tau}{\sigma\sqrt{\tau}}} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = e^{\frac{b^2 \sigma^2 \tau}{2}} \Phi\left(\frac{a}{\sigma\sqrt{\tau}} + b\sigma\sqrt{\tau}\right). \end{aligned}$$

□

Applying Lemma 3.1 to I_1 (with $a = x$, $b = -(1 - \alpha)$) and to I_2 (with $a = x$, $b = \alpha$), and substituting back to $u = e^{\alpha x + \beta \tau} v$, we get

$$\begin{aligned} I_1 &\Rightarrow I_3 = K \exp\left(\alpha x + \beta \tau + (1 - \alpha)x + \frac{(1 - \alpha)^2 \sigma^2 \tau}{2}\right) \Phi(d_1), \\ I_2 &\Rightarrow I_4 = -K \exp\left(\alpha x + \beta \tau - \alpha x + \frac{\alpha^2 \sigma^2 \tau}{2}\right) \Phi(d_2), \end{aligned}$$

where

$$\begin{aligned} d_1 &= \frac{x}{\sigma\sqrt{\tau}} + (1 - \alpha)\sigma\sqrt{\tau}, \\ d_2 &= \frac{x}{\sigma\sqrt{\tau}} - \alpha\sigma\sqrt{\tau} = d_1 - \sigma\sqrt{\tau}, \end{aligned}$$

rearranging and simplifying the expressions

$$\begin{aligned} I_3 &= K \exp\left(x + \left(\beta + \frac{(1 - \alpha)^2 \sigma^2}{2}\right) \tau\right) \Phi(d_1), \\ I_4 &= -K \exp\left(\left(\beta + \frac{\alpha^2 \sigma^2}{2}\right) \tau\right) \Phi(d_2), \end{aligned}$$

the exponent of the first term simplifies to

$$x + \left(\beta + \frac{(1-\alpha)^2 \sigma^2}{2} \right) \tau = x + \frac{1}{2\sigma^2} \left(- \left(r + \frac{1}{2} \sigma^2 \right)^2 + \left(r + \frac{1}{2} \sigma^2 \right)^2 \right) \tau = x,$$

and for the second one we use the fact that while deriving β , on the second equality we found

$$\beta = \frac{1}{2} \alpha \left(r - \frac{1}{2} \sigma^2 \right) - r$$

multiplying by $1 = \frac{\sigma^2}{\sigma^2}$ and rearranging we get

$$\beta = \frac{\sigma^2}{2} \alpha \frac{1}{\sigma^2} \left(r - \frac{1}{2} \sigma^2 \right) - r$$

and using the definition of α we get

$$\beta = -\frac{\sigma^2}{2} \alpha^2 - r.$$

Thus

$$\begin{aligned} I_3 &= K e^x \Phi(d_1), \\ I_4 &= -K e^{-r\tau} \Phi(d_2). \end{aligned}$$

Finally, reverting the substitutions $\tau = T - t$ and $x = \ln(S/K)$ we get the **Black-Scholes formula**

$$f(t, S) = S \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2), \quad (18)$$

where

$$\begin{aligned} d_1 &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + (1-\alpha)\sigma\sqrt{T-t} \\ &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + \left(1 + \frac{1}{\sigma^2} \left(r - \frac{1}{2}\sigma^2\right)\right) \sigma\sqrt{T-t} \\ &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + \left(\frac{1}{\sigma^2} \left(\sigma^2 + r - \frac{1}{2}\sigma^2\right)\right) \sigma\sqrt{T-t} \\ &= \frac{1}{\sigma\sqrt{T-t}} \left(\ln\left(\frac{S}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right) (T-t) \right), \end{aligned}$$

and, of course

$$\begin{aligned} d_2 &= d_1 - \sigma\sqrt{T-t} \\ &= \frac{1}{\sigma\sqrt{T-t}} \left(\ln\left(\frac{S}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right) (T-t) \right). \end{aligned}$$

3.4 Calibration - Implied volatility

The Black-Scholes formula (18) gives the call price as a function of market observables (t, S, r) , contract parameters (K, T) , and the single volatility parameter σ . In practice, market prices $C^{\text{mkt}}(K, T)$ are observable for a range of strikes and maturities, and the question is: which value of σ is implied by each observed price?

Definition 3.3 (Implied volatility). Given a market price C^{mkt} for a European call with strike K and maturity T , the **implied volatility** $\hat{\sigma}(K, T)$ is the unique $\sigma > 0$ such that

$$C_{\text{BS}}(t, S; K, T, \hat{\sigma}) = C^{\text{mkt}}, \quad (19)$$

where C_{BS} denotes the Black-Scholes formula (18).

Uniqueness follows from strict monotonicity of the call price in σ . To see this we compute the *vega* of the option, $\mathcal{V} := \partial C_{\text{BS}} / \partial \sigma$.

Write $\tau = T - t$ for brevity and note that d_1 and d_2 both depend on σ . Splitting d_1 into its two terms makes differentiation straightforward,

$$d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau},$$

so

$$\frac{\partial d_1}{\partial \sigma} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} + \frac{\sqrt{\tau}}{2}. \quad (20)$$

Since $d_2 = d_1 - \sigma\sqrt{\tau}$,

$$\frac{\partial d_2}{\partial \sigma} = \frac{\partial d_1}{\partial \sigma} - \sqrt{\tau} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} - \frac{\sqrt{\tau}}{2}. \quad (21)$$

In particular,

$$\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma} = \sqrt{\tau}. \quad (22)$$

Differentiating (18) with the chain rule gives

$$\frac{\partial C_{\text{BS}}}{\partial \sigma} = S \phi(d_1) \frac{\partial d_1}{\partial \sigma} - K e^{-r\tau} \phi(d_2) \frac{\partial d_2}{\partial \sigma},$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard Gaussian density. The two terms simplify if we use the identity

$$S \phi(d_1) = K e^{-r\tau} \phi(d_2). \quad (23)$$

To verify it, note that $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = 2(\ln(S/K) + r\tau)/(\sigma\sqrt{\tau})$, so

$$d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = 2\ln(S/K) + 2r\tau.$$

Therefore, since the $1/\sqrt{2\pi}$ factors cancel in the ratio,

$$\frac{\phi(d_2)}{\phi(d_1)} = \frac{e^{-d_2^2/2}}{e^{-d_1^2/2}} = e^{(d_1^2 - d_2^2)/2} = \exp\left(\ln \frac{S}{K} + r\tau\right) = \frac{S}{K} e^{r\tau},$$

which is exactly (23). Applying it and using (22),

$$\mathcal{V} := \frac{\partial C_{\text{BS}}}{\partial \sigma} = S \phi(d_1) \left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma} \right) = S \sqrt{T - t} \phi(d_1) > 0. \quad (24)$$

Since $\mathcal{V} > 0$, the map $\sigma \mapsto C_{\text{BS}}(\sigma)$ is a strictly increasing bijection from $(0, \infty)$ onto the no-arbitrage interval $(\max(S - K e^{-r(T-t)}, 0), S)$, so the implied volatility exists and is unique for any market price in this range.

Since (19) has no closed-form solution in σ , the implied volatility is computed numerically. The smoothness and strict positivity of $\mathcal{V}$ make Newton's method an efficient choice: starting from an initial guess $\sigma^{(0)} > 0$, one iterates

$$\sigma^{(n+1)} = \sigma^{(n)} - \frac{C_{\text{BS}}(\sigma^{(n)}) - C^{\text{mkt}}}{\mathcal{V}(\sigma^{(n)})}, \quad (25)$$

until $|C_{\text{BS}}(\sigma^{(n)}) - C^{\text{mkt}}| < \varepsilon$.

3.4.1 The implied volatility surface

If Black-Scholes with a single σ perfectly described the market, every option on the same underlying would yield the same implied volatility. In practice, the map $(K, T) \mapsto \hat{\sigma}(K, T)$ is non-trivial, and the resulting object is called the **implied volatility surface**.

Two features of this surface are well documented in the literature, see [4]:

- **Volatility skew:** for a fixed maturity T , equity index options exhibit a pronounced *negative skew*: implied volatility increases as the strike decreases relative to the spot. This reflects the demand for downside protection and the asymmetry of return distributions.
- **Term structure:** for a fixed strike, implied volatility varies with maturity, reflecting the market's expectation of future realized volatility at different horizons.

The non-flatness of the surface is direct empirical evidence that constant- σ Black-Scholes is inconsistent with observed market prices. This motivates more flexible models such as the *local volatility model*, which replaces σ by a deterministic function $\sigma(t, S)$ calibrated to reproduce the full surface exactly, as described in the following section.

Figure 3 shows the implied volatility surface computed from SPY option quotes on 2026-04-30, using the Black-Scholes-Merton formula with a continuously compounded dividend yield calibrated per maturity from put-call parity (see Appendix A for implementation details). The characteristic features are clearly visible: the volatility skew in the moneyness direction and the term structure in the maturity direction. Figure 4 shows the smile slice at the median maturity ($T \approx 0.21$ yr), where OTM puts (red) and OTM calls (blue) meet smoothly at the money.

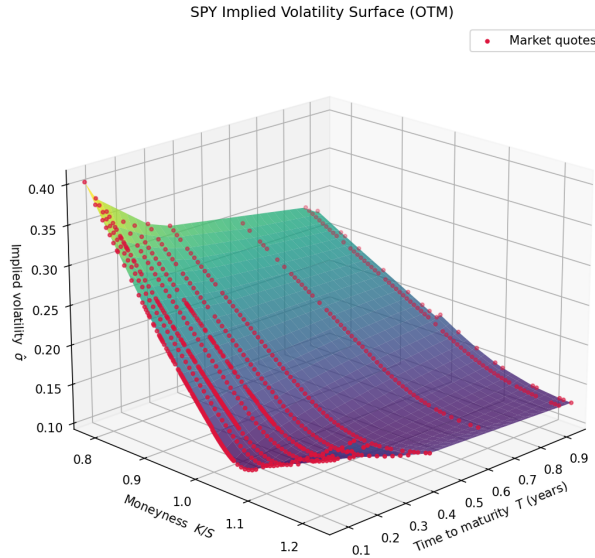


Figure 3: SPY implied volatility surface (OTM composite) on 2026-04-30. Axes: time to maturity T (years), moneyness K/S (inverted scale), implied volatility $\hat{\sigma}$. Red dots are individual market quotes; the surface is linearly interpolated.

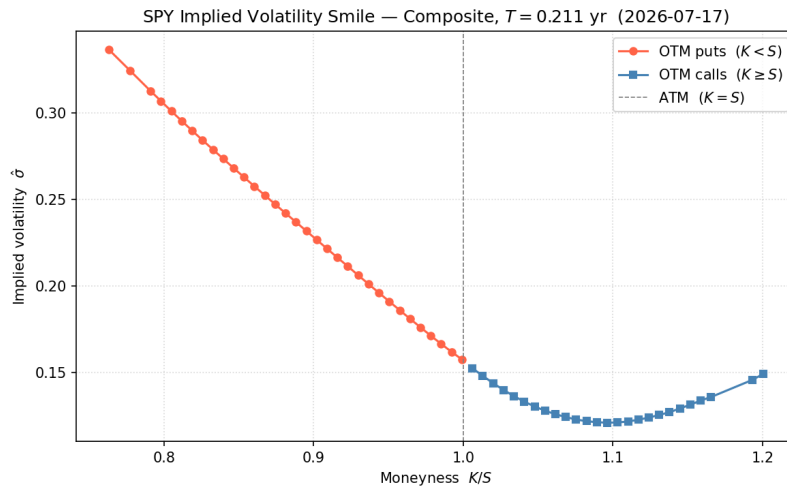


Figure 4: Volatility smile at the median maturity ($T \approx 0.21$ yr, expiration 2026-07-17). OTM puts (red) and OTM calls (blue) are joined from the calibrated composite surface. The smooth join at $K = S$ reflects the consistent dividend yield calibration.

4 Arbitrage theory

We can generalize the approach used to derive the Black-Scholes PDE for a more general products. Thus in this section we will try to give a general framework to derive equations to price a *contingent claim*.

Definition 4.1 (Market). A market is a set of $n + 1$ assets $S_0, S_1, \dots, S_n$, where the price of S_0 is given by

$$\frac{dS_0(t)}{S_0} = r(S_0(t), t)dt,$$

and the prices of the other assets are given by

$$dS_i(t) = \mu(S_i(t), t)dt + \sigma_i(S_i(t), t)^\top dW(t),$$

where $W(t)$ is a k -dimensional Brownian motion.

Definition 4.2 (Portfolio). A portfolio in the market $\{S_i(t)\}$ is an $n + 1$ dimensional $\mathcal{F}_t^{(k)}$ -adapted stochastic process

$$\theta(t) = (\theta_0(t), \theta_1(t), \dots, \theta_n(t)),$$

and its value at time t is given by

$$V(t) = \sum_{i=0}^n \theta_i(t) S_i(t).$$

Note that as a portfolio $\theta(t)$ is an adapted process it doesn't use future information, so it is a generalization on what in reality would be to rebalance one's investments in the market at time t .

We can generalize the self-financing condition presented in (14) as,

$$V(t) = V(0) + \int_0^t \theta(u) dS(u), \quad (26)$$

in differential form

$$dV(t) = \theta(t) dS(t). \quad (27)$$

A contingent claim is a financial instrument whose payoff depends on the outcome of uncertain future events, for example an European option. As our market consists of assets that are described as stochastic processes, an European contingent claim paying at time T is a function $f(S(T), T)$ depending on these assets, which is a random variable as $S(T)$ is a random vector of the values of the assets at time T .

4.1 PDE method

Consider an European contingent claim with payoff $f(S(T))$, similar to what we did in section 3.2, we want to construct a replicating portfolio $V(S(t), t)$ that has the same payoff as the contingent claim at time T . Thus using the Ito formula (5) we have

$$\begin{aligned} dV(S(t), t) &= \frac{\partial V}{\partial t} dt + \sum_{i=0}^n \frac{\partial V}{\partial S_i} dS_i(t) + \frac{1}{2} \sum_{i,j} \frac{\partial^2 V}{\partial S_i \partial S_j} dS_i(t) dS_j(t) \\ &= \sum_{i=0}^n \frac{\partial V}{\partial S_i} dS_i(t) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial^2 V}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\top \sigma_j(S(t), t) \right) dt, \end{aligned}$$

imposing the self-financing condition (27) we find that

$$\theta_i(t) = \frac{\partial V}{\partial S_i}, \quad \forall i = 0, \dots, n, \quad (28)$$

and

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial^2 V}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\top \sigma_j(S(t), t) = 0. \quad (29)$$

Also using $V(S(t), t) = \theta(t)S(t)$ in (28) we get

$$V(S(t), t) = \sum_{i=0}^n \frac{\partial V}{\partial S_i} S_i. \quad (30)$$

The equations (29) and (30) describe the PDE for the price of the contingent claim, using as boundary condition the payoff at time T

$$V(S(T), T) = f(S(T)).$$

As in Black-Scholes model, μ does not explicitly appear in the PDE, its only present implicitly in the prices $S(t)$ of the assets.

4.2 Martingale method

The martingale method is an alternative framework to the PDE method, that allows to price contingent claims in a more flexible way.

We will use chapter 10 of [2] as a reference for this section, we start with a couple of definitions related with the concept of market and portfolio defined previously.

Definition 4.3 (Attainable process). We say a price process $V(t)$ is *attainable* if there exists a self-financing portfolio $\theta(t)$ such that $V(t) = \theta(t)S(t)$.

The attainable process is a formalization of the dynamic replicating portfolio idea. An other restriction on the portfolios we are going to consider is the following,

Definition 4.4 (Admissible portfolio). A portfolio $\theta(t)$ is called *admissible* if exists $\alpha > 0$ such that

$$\int_0^t \theta(t) dS(t) \geq -\alpha, \quad \forall t \in [0, T].$$

In real life we can interpret this condition as a limit on the amount the portfolio can lose or borrow, as bank, clearing house or counterparty will not allow such risks and would force to cancel the deal. Technically also allow to cancel out some other undesirable strategies that would prevent some key theorems to hold, see [12] sections 14.5 and 14.6 for more details on such strategies.

Now we will formalize the idea of arbitrage in a market.

Definition 4.5 (Arbitrage). A self-financing and admissible portfolio $\theta(t)$ is an *arbitrage* if one of the following conditions hold for time t :

1. $\theta(0)S(0) < 0$ and $P(\theta(t)S(t) \geq 0) = 1$,
2. $\theta(0)S(0) = 0$ and $P(\theta(t)S(t) \geq 0) = 1$ and $P(\theta(t)S(t) > 0) > 0$,

Consider that the *risk-free* asset S_0 has a constant price of 1, i.e. the *risk-free* rate is $r(S(t), t) \equiv 0$. Under this assumption we introduce the *risk-neutral measure* $\mathcal{Q}$ as,

Definition 4.6. A probability measure $\mathcal{Q}$ on $\mathcal{F}_T$ is called a *risk-neutral measure* (also known as *equivalent martingale measure*) if

- $\mathcal{Q}$ is equivalent to P on $\mathcal{F}_T$.
- All price process in the market $S_0, S_1, \dots, S_n$ are martingales under $\mathcal{Q}$ on the time interval $[0, T]$.

We refer to P as the "real" probability measure and reflects the true probability of the price processes, whereas $\mathcal{Q}$ is a probability measure under which the expected price of any asset is equal to the current one regardless of their risk (volatility).

Under this assumption we have the following theorem.

Theorem 4.1 (First fundamental theorem of asset pricing). The market $\{S_i(t)\}$ is arbitrage-free if and only if there exists an equivalent (local) martingale measure $\mathcal{Q}$.

We include the local martingales as restricting to the martingales would eliminate too many processes, that are of interest.

The restriction on the risk-free asset being constant can be dropped by considering the first price process a numeraire and normalizing the market. For S_0 to be a numeraire we need that $S_0(t) > 0$ almost surely for all $t \geq 0$. Using S_0 as numeraire we can consider the processes $Z_i = S_i(t)/S_0(t)$, which are the *discounted* price processes of the assets.

Definition 4.7 (Normalized market). A normalized market is a vector of processes Z given by

$$Z_i(t) = \frac{S_i(t)}{S_0(t)}, \quad i = 1, \dots, n.$$

The results we have presented so far are invariant to the usage of the "regular" market $\{S_i(t)\}$ as defined in (4.1) or the normalized market $\{Z_i(t)\}$ (4.7). So if a portfolio is self-financing on $\{S_i(t)\}$ then it is also self-financing on $\{Z_i(t)\}$ it is also self-financing on $\{Z_i(t)\}$, and if a portfolio does not have arbitrage on $\{S_i(t)\}$ then it is also self-financing on $\{Z_i(t)\}$ it does not have it on $\{Z_i(t)\}$ either. Thus 4.1 holds for any market $\{S_i(t)\}$ by just normalizing it.

Notice that if the numeraire is the risk-free asset S_0 then the the market $\{S_i(t)\}$ is arbitrage-free if all assets $S_i(t)$ have a local rate of return $r(t)$ under $\mathcal{Q}$, that is, if $\mu_i(S_i(t), t) = r(t)S_i(t)$ for all $i = 1, \dots, n$ under $\mathcal{Q}$.

Under the assumption of a market with no arbitrage, a key question is whether a contingent claim can be replicated by a portfolio consisting on the market assets. This yields the following theorem.

Theorem 4.2 (Second fundamental theorem of asset pricing). Assuming absence of arbitrage, the market model is complete if and only if the martingale measure $\mathcal{Q}$ is unique.

Finally we can give a general formula to price contingent claims. Under the assumption of a complete market, the price of a contingent claim $f(S(T))$ is given by

$$V(t, f(S(T))) = \mathbb{E}^{\mathcal{Q}} S_0(t) \left[\frac{f(S(T))}{S_0(T)} | \mathcal{F}_t \right] \quad (31)$$

in case that the numeraire is the risk-free asset,

$$V(t, f(S(T))) = \mathbb{E}^{\mathcal{Q}} \left[e^{\int_t^T r(s) ds} f(S(T)) | \mathcal{F}_t \right]$$

5 Local Volatility Model

We are going to introduce the Local Volatility Model mainly following [1].

5.1 Model description

The **local volatility model** is the simplest and most widely used volatility model in derivatives pricing. It extends the Black-Scholes framework by allowing the instantaneous volatility to be a deterministic function of both time and the spot price of the underlying asset,

$$\sigma = \sigma(t, S).$$

So the dynamics of the underlying asset under the local volatility model are given by the following stochastic differential equation,

$$\frac{dS_t}{S_t} = (r - q) dt + \sigma(t, S_t) dW_t,$$

and the pricing PDE is identical to the Black-Scholes PDE, with the deterministic volatility function replacing the constant volatility

$$\frac{\partial f}{\partial t} = rf(t, S_t) - (r - q)S_t \frac{\partial f}{\partial S} - \frac{1}{2}\sigma^2(t, S_t)S_t^2 \frac{\partial^2 f}{\partial S^2},$$

5.1.1 Market model

A market model is a valuation model that includes vanilla options as part of the underlying, this means we are assuming that when hedging a derivative, we can use the vanilla options as a hedging instrument, in other words, the derivative can be replicated using a combination of the underlying as well as vanilla options. Thus vanilla options under a market model are exactly priced.

The local volatility model is a market model, since it is calibrated to match the vanilla option prices observed in the market

5.2 Dupire Equation

The Dupire equation, introduced in [5], expresses the local volatility function $\sigma(t, S)$ directly in terms of derivatives of call option prices with respect to strike and maturity. We present the derivation following [1].

5.2.1 Derivation

Consider the following general diffusive dynamics for S_t :

$$dS_t = (r - q) S_t dt + \sigma_t S_t dW_t, \tag{32}$$

where σ_t is an arbitrary adapted process. The price of a European call option with strike K and maturity T under the risk-neutral measure is:

$$C(K, T) = e^{-rT} \mathbb{E}[(S_T - K)^+].$$

Before applying Itô's formula we recall the two generalized functions that appear as distributional derivatives of $(x)^+ = \max(x, 0)$.

Definition 5.1 (Heaviside function). The **Heaviside function** $\theta : \mathbb{R} \rightarrow \{0, 1\}$ is defined by

$$\theta(x) = \begin{cases} 1 & x > 0, \\ 0 & x < 0. \end{cases}$$

Definition 5.2 (Dirac delta). The **Dirac delta** δ is the distributional derivative of θ , characterized by the property that for any smooth test function φ :

$$\int_{-\infty}^{+\infty} \varphi(x) \delta(x) dx = \varphi(0).$$

These two distributions arise as the successive generalized derivatives of $(x)^+$:

$$\frac{d}{dx}(x)^+ = \theta(x), \quad \frac{d^2}{dx^2}(x)^+ = \delta(x). \quad (33)$$

The first identity follows from $(x)^+ = x \theta(x)$ and $x \delta(x) = 0$; the second is the distributional derivative of θ .

To relate option prices to the dynamics of S_t , we apply Itô's formula to $(S_T - K)^+$ over the interval $[T, T + dT]$, using (33) with $x = S_T - K$:

$$\begin{aligned} d(S_T - K)^+ &= \theta(S_T - K) \left((r - q) S_T dT + \sigma_T S_T dW_T \right) \\ &\quad + \frac{1}{2} \delta(S_T - K) \sigma_T^2 S_T^2 dT. \end{aligned}$$

We work with undiscounted prices $\mathcal{C}(K, T) = e^{rT} C(K, T)$. Differentiating the identity $\mathcal{C}(K, T) = \mathbb{E}[(S_T - K)^+]$ with respect to K gives:

$$\mathbb{E}[\theta(S_T - K)] = -\frac{\partial \mathcal{C}}{\partial K}, \quad \mathbb{E}[\delta(S_T - K)] = \frac{\partial^2 \mathcal{C}}{\partial K^2}. \quad (34)$$

The second identity is the Breeden-Litzenberger result [3]: the second strike derivative of undiscounted call prices equals the risk-neutral density of S_T . To see why, let $p(\cdot, T)$ denote the risk-neutral density of S_T , so that

$$\mathcal{C}(K, T) = \mathbb{E}[(S_T - K)^+] = \int_0^\infty (s - K)^+ p(s, T) ds = \int_K^\infty (s - K) p(s, T) ds.$$

Differentiating with respect to K (using Leibniz's rule, noting the lower limit contributes zero because $(K - K)^+ = 0$):

$$\frac{\partial \mathcal{C}}{\partial K} = - \int_K^\infty p(s, T) ds = -(1 - F(K, T)),$$

where $F(K, T) = P(S_T \leq K)$ is the risk-neutral CDF. A second differentiation gives:

$$\frac{\partial^2 \mathcal{C}}{\partial K^2} = p(K, T). \quad (35)$$

Since $\mathbb{E}[\delta(S_T - K)] = \int_0^\infty \delta(s - K) p(s, T) ds = p(K, T)$, this is exactly the second identity in (34). In particular, the positivity of p guarantees $\partial^2 \mathcal{C} / \partial K^2 \geq 0$, which we will use when discussing no-arbitrage conditions.

From the identity $\mathcal{C} = \mathbb{E}[S_T \theta(S_T - K)] - K \mathbb{E}[\theta(S_T - K)]$ we also obtain:

$$\mathbb{E}[S_T \theta(S_T - K)] = \mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K}.$$

Taking the expectation of the Itô expansion (noting $\mathbb{E}[dW_T] = 0$) and substituting these identities:

$$\begin{aligned} \frac{\partial \mathcal{C}}{\partial T} dT &= (r - q) \mathbb{E}[S_T \theta(S_T - K)] dT + \frac{1}{2} \mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] dT \\ &= (r - q) \left(\mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K} \right) dT + \frac{1}{2} \mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] dT. \end{aligned}$$

For the remaining term, we use the sifting property of the Dirac delta: since $\delta(S_T - K)$ is supported at $S_T = K$, the factor S_T^2 can be replaced by K^2 inside any expectation,

$$\mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] = K^2 \mathbb{E}[\sigma_T^2 \delta(S_T - K)],$$

giving:

$$\frac{\partial \mathcal{C}}{\partial T} dT = (r - q) \left(\mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K} \right) dT + \frac{K^2}{2} \mathbb{E}[\sigma_T^2 \delta(S_T - K)] dT.$$

Solving for $\mathbb{E}[\sigma_T^2 \delta(S_T - K)]$, dividing by $\mathbb{E}[\delta(S_T - K)] = \partial^2 \mathcal{C} / \partial K^2$, and reverting to discounted prices $C = e^{-rT} \mathcal{C}$, we obtain:

$$\mathbb{E}[\sigma_T^2 | S_T = K] = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q) K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}}. \quad (36)$$

This identity holds for any model of the lognormal-type form 32, with σ_t an arbitrary adapted process [1]. The multiplicative structure $\sigma_t S_t$ is essential: it is what produces the factor S_T^2 in the Itô expansion, which the sifting property then converts to K^2 . A consequence is that the smile alone does not determine σ_t : two processes σ_t and σ'_t generate the same vanilla smile if and only if $\mathbb{E}[\sigma_T^2 | S_T = K] = \mathbb{E}[\sigma_T'^2 | S_T = K]$ for all K and T .

In the local volatility model, where $\sigma_t \equiv \sigma(t, S_t)$, the conditional expectation reduces to $\sigma(t, S)^2$, and evaluating at $K = S$, $T = t$ gives the **Dupire formula**:

$$\sigma(t, S)^2 = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q) K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}} \Bigg|_{\substack{K=S, \\ T=t}}. \quad (37)$$

Given the full surface of observed call prices $C(K, T)$, equation (37) uniquely determines the local volatility function.

5.2.2 Forward equation

Rearranging (36) for a known local volatility function $\sigma(t, S)$ yields the **Dupire forward equation**:

$$\frac{\partial C}{\partial T} + (r - q) K \frac{\partial C}{\partial K} - \frac{\sigma^2(T, K)}{2} K^2 \frac{\partial^2 C}{\partial K^2} = -qC. \quad (38)$$

Unlike the Black-Scholes backward PDE, which is solved backwards in t from a terminal payoff for a single strike and maturity, the forward equation propagates forward in T with initial condition $C(K, T = 0) = (S_0 - K)^+$, producing prices for a fixed initial spot S_0 but all strikes K and maturities T simultaneously.

5.2.3 No-arbitrage conditions

For the Dupire formula (37) to yield a positive local variance, both the numerator and denominator of (36) must be strictly positive, which is equivalent to requiring the option surface to be free of arbitrage.

Strike arbitrage. From (34), the denominator satisfies:

$$\frac{\partial^2 C(K, T)}{\partial K^2} = e^{-rT} \mathbb{E}[\delta(S_T - K)] \geq 0,$$

since it is proportional to the risk-neutral density of S_T . A violation can be exploited via a *butterfly spread*: buying ε^{-2} calls at strike $K - \varepsilon$, buying ε^{-2} calls at strike $K + \varepsilon$, and selling $2\varepsilon^{-2}$ calls at strike K . This portfolio has a non-negative payoff at maturity for any realisation of S_T , so its inception value must be non-negative, implying $\partial^2 C / \partial K^2 \geq 0$.

Maturity arbitrage. The numerator of (36) can be rewritten as:

$$e^{-qT} \frac{d}{dT} \left[e^{qT} C \left(K e^{(r-q)T}, T \right) \right],$$

which is non-negative if and only if e^{qT} times the price of a call struck at a fixed proportion of the forward $F_T = S_0 e^{(r-q)T}$ is non-decreasing in maturity. A violation constitutes a *maturity arbitrage*: one can buy a longer-dated option and sell a shorter-dated one at a net positive premium, with a delta position that generates non-negative P&L at both maturities.

In an arbitrage-free market both conditions hold and the Dupire local volatility function is well-defined.

5.3 Calibration problem

The local volatility model must be calibrated to market option prices to be usable. Given quoted call prices $C^*(K_i, T_j)$ for a finite set of strikes and maturities, we seek a function $\sigma(S, t)$, or equivalently $a(y, \tau) = \frac{1}{2}\sigma^2$, such that the Dupire forward equation reproduces the observed prices. This is an ill-posed inverse problem: market data are noisy, sparse in maturity, and the map from a to option prices has a smoothing effect that makes its inverse discontinuous. We follow [6] and treat the calibration as a Tikhonov regularization problem.

5.4 Tikhonov regularization of the inverse pricing problem

5.4.1 Change of variable - log-moneyness

Market option prices depend on the ratio K/S_0 rather than on the absolute strike K . Working in log-moneyness

$$y = \log(K/S_0), \quad \text{equivalently} \quad K = S_0 e^y,$$

removes this dependence and puts the PDE in a translation-invariant form, just as the substitution $x = \log(S/K)$ reduced the Black-Scholes PDE to the heat equation in Sec-

tion 3.3. Define the time to maturity $\tau = T$, the diffusion coefficient $a(y, \tau) = \frac{1}{2}\sigma^2(K, T)$, and the normalized call price $u(y, \tau) = C(S_0; K, T)/S_0$, so that $C(S_0; K, T) = S_0 u(y, \tau)$.¹

We now carry the Dupire forward equation (38) into the (y, τ) coordinates. Since $y = \log(K/S_0)$ we have $\partial y/\partial K = 1/K$, and because $\tau = T$ the maturity derivative is unchanged. Applying the chain rule to $C = S_0 u(y, \tau)$ gives the three derivatives that appear in (38):

$$\begin{aligned}\frac{\partial C}{\partial T} &= S_0 u_\tau, \\ \frac{\partial C}{\partial K} &= S_0 u_y \frac{\partial y}{\partial K} = \frac{S_0}{K} u_y, \\ \frac{\partial^2 C}{\partial K^2} &= \frac{\partial}{\partial K} \left(\frac{S_0}{K} u_y \right) = \frac{S_0}{K} u_{yy} \frac{\partial y}{\partial K} - \frac{S_0}{K^2} u_y = \frac{S_0}{K^2} (u_{yy} - u_y).\end{aligned}$$

The convenient feature of log-moneyness is that the first-order operator $K \partial_K$ and the second-order operator $K^2 \partial_{KK}$ both become constant-coefficient operators in y :

$$K \frac{\partial C}{\partial K} = S_0 u_y, \quad K^2 \frac{\partial^2 C}{\partial K^2} = S_0 (u_{yy} - u_y),$$

which is exactly what makes the equation translation-invariant in y . Substituting these into (38) and using $a = \frac{1}{2}\sigma^2$,

$$S_0 u_\tau + (r - q) S_0 u_y - a S_0 (u_{yy} - u_y) = -q S_0 u.$$

Dividing through by S_0 and solving for u_τ yields the forward equation in log-moneyness form,

$$u_\tau = a(y, \tau) (u_{yy} - u_y) - (r - q) u_y - q u, \quad (39)$$

on the truncated domain $Q_{M,T} = (-M, M) \times (0, T]$. The initial condition follows from the same normalization applied to the call payoff $C(K, 0) = (S_0 - K)^+$:

$$u(y, 0) = \frac{(S_0 - K)^+}{S_0} = \left(1 - \frac{K}{S_0}\right)^+ = (1 - e^y)^+,$$

that is,

$$u(y, 0) = \max(1 - e^y, 0) \quad (40)$$

and Dirichlet boundary conditions

$$u(-M, \tau) = \Psi_1(\tau), \quad u(M, \tau) = \Psi_2(\tau). \quad (41)$$

The boundary values are determined from the Black-Scholes formula evaluated at the boundary strikes $K = S_0 e^{\pm M}$: as $M \rightarrow \infty$, the left boundary ($K \rightarrow 0$) gives the intrinsic value $\Psi_1(\tau) \rightarrow e^{-q\tau}$, and the right boundary ($K \rightarrow \infty$) gives $\Psi_2(\tau) \rightarrow 0$.

¹We keep the explicit zero-order term. [6] instead set $u = e^{\int_t^T q ds} C$, which absorbs the dividend factor and yields the equivalent equation $-u_\tau + a(u_{yy} - u_y) + (q - r)u_y = 0$ without a $-qu$ term; the two normalizations coincide when $q = 0$, as in Example 1 below.

5.4.2 Parameter-to-solution map

For each admissible diffusion coefficient a , let $u(a)$ denote the solution to (39)–(40). Fix an *a priori* reference $a^* \in L^\infty(\mathbb{R})$ with $0 < \underline{a} \leq a^* \leq \bar{a}$ and define the set of admissible parameters

$$\mathcal{K}(a^*) = \{a \in a^* + H^1(Q) : \underline{a} \leq a \leq \bar{a}\}. \quad (42)$$

Here $Q = \Omega \times (0, T)$ is the space-time domain and $a^* + H^1(Q)$ is taken with $\nabla a^* \in [L^2(Q)]^2$ (Definition 2.1 of [6]). Two features of H^1 matter. The coefficient enters the equation through the flux term of the divergence form (50), so it must have a weak derivative in L^2 . And the H^1 penalty makes the sublevel sets of the Tikhonov functional bounded in H^1 , hence weakly compact, which is what yields a minimizer. In the numerical setting below the volatility is time independent, $a = a(y)$, and the penalty reduces to the spatial norm $\int_{-M}^M (a^2 + a_y^2) dy$; on the bounded interval H^1 then also embeds into $C[-M, M]$, so the bounds $\underline{a} \leq a \leq \bar{a}$ hold pointwise.

The *parameter-to-solution map* $F = F_{a_0}$ is defined as

$$F(a) = u(a) - u(a_0), \quad (43)$$

where $a_0 \in \mathbb{R}$ with $\underline{a} \leq a_0 \leq \bar{a}$ is a reference constant and $u(a_0)$ can be computed from the Black-Scholes formula. A key regularity result established in [6] is the following.

Theorem 5.1 ([6], Theorem 2.1). There exists $p^* > 2$ such that $F: \mathcal{K}(a^*) \rightarrow W_p^{2,1}(Q)$ is continuous and compact for $2 \leq p < p^*$, and weakly sequentially continuous.

The compactness of F is the mathematical source of the ill-posedness. It reflects the smoothing action of the pricing equation: the solution of a parabolic problem is more regular than its coefficient, and the solution space $W_p^{2,1}(Q)$ embeds compactly in the observation space. A compact operator between infinite dimensional spaces cannot have a continuous inverse, so arbitrarily small changes in the observed prices can correspond to order one changes in a . In practice very different volatility profiles reproduce almost the same option prices, especially away from the money, and any direct inversion amplifies the noise. Controlling this instability is the purpose of the Tikhonov penalty.

5.4.3 Tikhonov functional

Let u^δ denote the (noisy) observed option prices with $\|u^\delta - u^\dagger\|_{L_2(Q)} \leq \delta$, where $u^\dagger = u(a^\dagger)$ for the true (unknown) parameter $a^\dagger$. Following [6] we stabilize the inverse problem by minimizing the *Tikhonov functional*

$$J^\beta(a) = \|u(a) - u^\delta\|_{L_2}^2 + \beta \|a - a^*\|_{H^1}^2, \quad (44)$$

where $\beta > 0$ is the regularization parameter, u^δ is the observation (option prices at a fixed maturity T_0), and a^* is the *a priori* guess that encodes any structural knowledge about the volatility (e.g., smoothness or decay for large $|y|$). The first term penalizes mismatch with data; the second term penalizes deviation of a from a^* in H^1 , which promotes smooth reconstructions.

The H^1 norm in the penalty is

$$\|a - a^*\|_{H^1}^2 = \int_{-M}^M [(a - a^*)^2 + (\partial_y(a - a^*))^2] dy.$$

Proposition 5.2 ([6], Propositions 3.1 and 3.2). For any $\beta > 0$ a minimizer a_β^δ of J^β exists. If $\delta \rightarrow 0$ and $\beta(\delta) \rightarrow 0$ with $\delta^2/\beta(\delta) \rightarrow 0$, then every sequence of minimizers $\{a_{\beta(\delta_k)}^{\delta_k}\}$ has a convergent subsequence, and every limit point is an a^* -minimum-norm solution.

Idea of the proof. Both statements are the standard results for Tikhonov regularization of nonlinear ill-posed problems [7], which Theorem 2.1 makes applicable here. For existence, a minimizing sequence $\{a_n\}$ is bounded in H^1 by the penalty term, so it has a weakly convergent subsequence $a_n \rightharpoonup \bar{a}$. The constraint set is convex and closed, hence weakly closed, so $\bar{a} \in \mathcal{K}(a^*)$. Weak sequential continuity of F (Theorem 2.1) and weak lower semicontinuity of the H^1 norm give $J^\beta(\bar{a}) \leq \liminf_n J^\beta(a_n)$, so $\bar{a}$ is a minimizer. For the vanishing noise limit, the choice $\delta^2/\beta(\delta) \rightarrow 0$ keeps the data and penalty terms balanced, so the minimizers stay bounded and every weak limit point solves the noise free problem with least H^1 distance to a^* . We refer to [7] for the general theory and to [6] for its application here. $\square$

5.4.4 Gradient computation via the adjoint method

To minimize J^β by gradient-based methods we need its Gâteaux derivative. The *directional derivative* of the data-fidelity term at a in direction h is

$$\frac{d}{d\varepsilon} \|u(a + \varepsilon h) - u^\delta\|_{L_2}^2 \Big|_{\varepsilon=0} = 2(u(a) - u^\delta, u'(a)[h])_{L_2},$$

where $u'(a)[h] = w(\cdot, T_0)$ is the solution of the *linearized* forward equation, obtained by differentiating (39) in the direction h ,

$$-w_\tau + a(w_{yy} - w_y) - (r - q)w_y - qw = -h(u_{yy} - u_y), \quad w(y, 0) = 0, \quad w|_{\partial\Omega} = 0. \quad (45)$$

Rather than solving (45) for every direction h , the gradient is evaluated efficiently via a single *adjoint solve*. Let $V(y, \tau)$ solve the backward parabolic problem

$$V_\tau + (aV)_{yy} + (aV)_y + (r - q)V_y - qV = 0, \quad V(y, T_0) = u(a)(y, T_0) - u^\delta(y), \quad (46)$$

with homogeneous Dirichlet conditions at $y = \pm M$ and integrated backward in τ . Then the Gâteaux derivative satisfies [6]

$$(u'(a)[h], u(a) - u^\delta)_{L_2} = \int_0^{T_0} \int_{-M}^M h(y) (u_{yy} - u_y) V dy d\tau.$$

Adding the H^1 regularization term, the total gradient direction (in L^2 sense) is

$$\langle \nabla J^\beta(a), h \rangle = 2 \int_0^{T_0} \int_{-M}^M h (u_{yy} - u_y) V dy d\tau + 2\beta \int_{-M}^M [(a - a^*)h + \partial_y(a - a^*)\partial_y h] dy. \quad (47)$$

In the finite-element implementation the data term is assembled in the equivalent form obtained by integrating $(u_{yy} - u_y)$ by parts against the basis functions, which avoids a mass-matrix solve and yields the Euclidean gradient directly. The second term is the H^1 inner product $(a - a^*, h)_{H^1}$, whose Euclidean (DOF-level) representation is $(M + K)(a - a^*)$, where M and K are the finite-element mass and stiffness matrices.

5.4.5 Convergence rates and the source condition

The adjoint solve of the previous section computes the action of $F'(a)^*$ on the residual. Writing the data misfit as $\|F(a) - u^\delta\|^2$ and using

$$\langle \nabla J_d(a), h \rangle = 2(F'(a)h, F(a) - u^\delta) = 2(h, F'(a)^*(F(a) - u^\delta)),$$

the field V of (46) represents $F'(a)^*(F(a) - u^\delta)$ (Remark 4.2 of [6]). The same adjoint controls the speed of convergence.

Convergence alone, as in the proposition above, gives no rate: without a further assumption it can be arbitrarily slow. The general theory [7] provides a rate under a source condition on the true coefficient,

$$a^* - a^\dagger = F'(a^\dagger)^* w, \quad (48)$$

read in the H^1 inner product, where $F'(a^\dagger)$ is the linearization $h \mapsto u'(a^\dagger)[h]$ of (45). With the differentiability and Lipschitz bounds on F established in Proposition 4.1 of [6] and $\|w\|$ small enough, the choice $\beta \sim \delta$ gives

$$\|a_\beta^\delta - a^\dagger\| = O(\sqrt{\delta}), \quad \|u(a_\beta^\delta) - u^\delta\| = O(\delta). \quad (49)$$

The abstract condition (48) is hard to check directly, so Egger and Engl derive interpretable sufficient conditions by restricting to a single maturity and a time independent volatility. In that setting they prove two rate results of the form (49). Theorem 4.1 treats observations of prices, with the penalty strengthened to the H^2 norm. Theorem 4.2 keeps the H^1 penalty but assumes that the first price derivative is also observed, which is realistic because $C_K = e^{-y}u_y$ is the price of a digital call and is quoted on the market. In both, the source function must be smooth and decay like a Gaussian in the wings: with $g = \partial_y^4 w - \partial_y^2 w + w$ for the first result and $g = -w_{yy} + w$ for the second, the requirement is $|g|, |g_y|, |g_{yy}| = O(e^{-B'y^2})$ as $|y| \rightarrow \infty$.

This decay condition is the precise form of the wing behaviour, and it explains the shape of a good prior. For strikes far from the money the prices carry almost no volatility information, so stable reconstruction there is impossible and the missing information has to come from a^* . The condition asks $a^\dagger - a^*$ to be small and smooth in the wings, which is why the priors in the experiments below are chosen to flatten away from the money. The effect is visible in Example 1, where the reconstruction returns to a^* in exactly the region the data cannot constrain.

Two points carry over to the rest of the chapter. The functional we minimize keeps the H^1 penalty with price data, which is the setting of the stability and convergence results but not literally of either rate theorem, since those need the stronger H^2 penalty or the added derivative data. And the rate analysis is proved only for a time independent volatility, so a genuinely time dependent surface $a(y, \tau)$ lies outside its scope.

5.4.6 Choice of β and stopping criterion

In the numerical experiments of [6] the regularization parameter is chosen by a decreasing sequence

$$\beta_0 = 10^2 \delta > \beta_1 > \dots > \beta_n = 10^{-3} \delta,$$

minimizing J^{β_i} sequentially, warm-starting each minimization from the previous minimizer. The sequence is stopped when the Morozov discrepancy principle $\|u(a_\beta^\delta) - u^\delta\| \leq \tau \delta$ is satisfied for some $\tau > 1$; across the noise levels $\delta \in [10^{-5}, 10^{-1}]$ tested in [6], the choice $\tau = 1.5$ is met at typical values $\beta \sim 10^{-2} \delta$. For exact data ($\delta = 0$) a single fixed β is used.

5.4.7 Variational forms and finite-element discretization

The forward PDE (39) is discretized in space with P_1 finite elements on a uniform mesh of $[-M, M]$ and in time with the implicit Euler scheme. Multiplying by a test function $v \in H_0^1(-M, M)$ and integrating by parts gives

$$\frac{d}{d\tau} \int_{-M}^M u v dy = - \int_{-M}^M a u_y v_y dy - \int_{-M}^M \partial_y a u_y v dy - \int_{-M}^M (a+r-q) u_y v dy - \int_{-M}^M q u v dy. \quad (50)$$

The adjoint equation (46) is run backward in τ ; its weak form (with $m = V$, $m_\tau = -(am)_{yy} - (am)_y - (r-q)m_y + qm$) is

$$\frac{d}{d\tau} \int_{-M}^M m v dy = \int_{-M}^M (am)_y v_y dy + \int_{-M}^M a m v_y dy + \int_{-M}^M (r-q) m v_y dy + \int_{-M}^M q m v dy. \quad (51)$$

Both systems are solved with a direct LU factorization at each time step. Gradient updates are assembled from (47) using the stored forward trajectory and the backward adjoint solution.

5.4.8 Verification of the discrete adjoint gradient

The gradient (47) is assembled from the discrete forward and adjoint solutions rather than by differentiating the discrete functional in closed form. Before handing it to the optimizer we check that it is indeed the gradient of the discrete Tikhonov functional. We use two tests; a third, `scipy.optimize.check_grad`, returns a relative error of 3.7×10^{-7} .

Tangent-linear/adjoint duality. Write the data-fidelity term as $J_d(a) = \|u(a) - u^\delta\|_{L_2}^2$, with residual $r = u(a) - u^\delta$. The gradient returned by the adjoint solve should satisfy

$$\langle \nabla J_d(a), h \rangle_{\mathbb{R}^n} = 2(r, u'(a)[h])_{L_2}$$

for every direction h , with the right-hand side computed separately from the linearized solve (45). This equality is exact, with no finite-difference truncation, so if it holds the discrete adjoint is the exact transpose of the linearized forward operator. For a random h on the Example 1 grid the two sides differ by 2.3×10^{-15} , i.e. at the level of rounding error.

Taylor remainder test. When the gradient is correct, the first-order Taylor remainder of the full functional decays one order faster in t than the zeroth-order one:

$$R_0(t) = |J(a+th) - J(a)| = O(t), \quad R_1(t) = |J(a+th) - J(a) - t \langle \nabla J(a), h \rangle| = O(t^2).$$

Table 1 lists R_0 and R_1 as the step t is reduced geometrically. The remainder R_1 falls at rate ≈ 2 , as it should when ∇J is exact.

t	$R_0(t)$	$R_1(t)$	rate of R_1
10^{-2}	1.32×10^{-6}	5.13×10^{-6}	-
10^{-3}	3.28×10^{-7}	5.27×10^{-8}	1.99
10^{-4}	3.76×10^{-8}	5.29×10^{-10}	2.00
10^{-5}	3.80×10^{-9}	5.29×10^{-12}	2.00

Table 1: Taylor remainder test for the discrete adjoint gradient (Example 1 setup). The zeroth-order remainder R_0 decays at first order in t and the first-order remainder R_1 at second order (rate ≈ 2), which is what an exact gradient produces. The rate column is $\log_{10}(R_1(10t)/R_1(t))$.

5.4.9 Numerical experiment - Example 1

We reproduce Example 1 (case A) of [6]: constant volatility recovery on $y \in [-3, 3]$ ($M = 3$), $\tau \in [0, 1]$, with $r = q = 0$ and spot $S_0 = 1000$.² The true parameter is

$$a^\dagger = 0.15 \quad \Longleftrightarrow \quad \sigma^\dagger = \sqrt{2 \times 0.15} \approx 0.548.$$

Data u^δ are the exact option prices at $\tau = T = 1$ generated from the Black-Scholes formula (no noise, case A of the paper). The initial guess and a^* are both set to

$$a_1^*(y) = 0.15 - 0.05 \operatorname{erf}(-y^2),$$

which agrees with the true value at the money and flattens out ($\partial_y a^* \rightarrow 0$, so $a^* \rightarrow 0.20$) for large $|y|$, the asymptotic behaviour invoked in [6] for the convergence-rate theorems 4.1-4.2. The regularization parameter is fixed at $\beta = 10^{-6}$ (exact data, so a small noise level).

The PDE is discretized with 199 linear (P_1) elements on $[-M, M]$, clustered toward the money by a sinh node map, since the price data are informative only near $y = 0$, and 200 implicit-Euler steps in τ , and the Tikhonov functional is minimized by L-BFGS-B using the adjoint gradient (47). Figure 5 shows the result: the recovered $a(y)$ matches the true constant $a^\dagger = 0.15$ across the strike range the data constrain (reproducing Tables 1–2 of [6] to four digits), while in the deep in-/out-of-the-money tails, where the prices carry almost no volatility information, it reverts to the *a priori* guess a_1^* , a direct manifestation of the ill-posedness discussed above.

5.4.10 Numerical experiment - Example 2

Example 1 has a constant target and is recovered almost exactly, so it mainly exercises the forward and adjoint solvers. Example 2 of [6] puts more weight on the regularization. Its true coefficient is the oscillatory profile

$$a^\dagger(y) = 0.15 + 0.05 e^{-(y+0.3)^2} \sin(2\pi y) + 0.05 \operatorname{erf}(20y), \quad (52)$$

with a steep step at the money (the $\operatorname{erf}(20y)$ term) and oscillations that decay into the wings. The domain, rates and spot are those of Example 1 ($y \in [-3, 3]$, $r = q = 0$,

²The paper’s text states $S_0 = 100$, but its tabulated option values (e.g. a call worth 439 at $K = 600$) and test (D) (“an underlying worth 1000\$”) are only consistent with $S_0 = 1000$; we use the latter. The recovered $a(y)$ is scale-invariant, but the Tikhonov balance is not, the data term scales as S_0^2 while the penalty does not, so reproducing the paper’s β requires its S_0 .

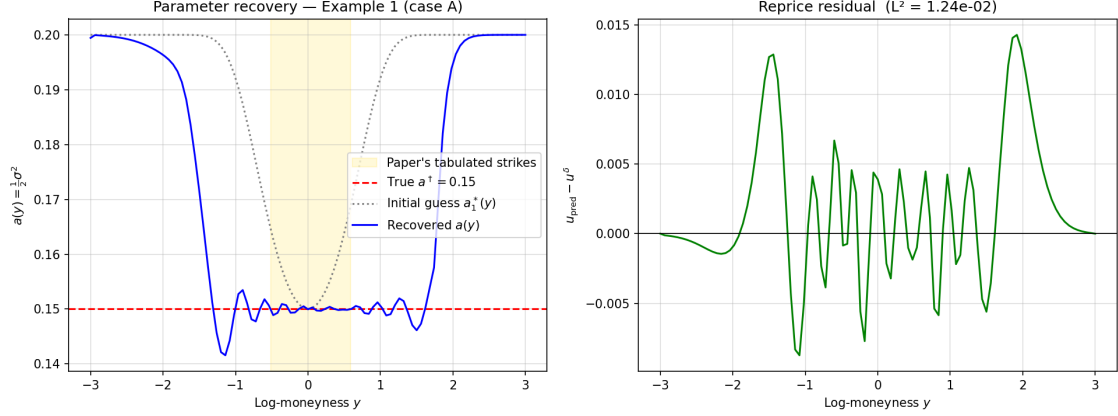


Figure 5: Example 1 (case A) of [6], constant-volatility recovery. *Left*: recovered $a(y) = \frac{1}{2}\sigma^2$ (blue) against the true value $a^\dagger = 0.15$ (red dashed) and the initial guess/prior a_1^* (grey dotted); the shaded band marks the strikes $K = 600\text{--}1800$ tabulated in the paper. *Right*: the repricing residual $u_{\text{pred}} - u^\delta$ at $\tau = T$.

$S_0 = 1000$, $\tau \in [0, 1]$). To avoid an *inverse crime* we generate the synthetic data on a wider domain $[-5, 5]$, with a four-times finer mesh and time step, and interpolate them onto the coarser reconstruction grid, so the data do not come from the same discretized operator that is later inverted; [6] use a different time-integration scheme for the same reason.

We run the paper's five cases. All use the *a priori* guess $a_1^*(y) = 0.15 + 0.05 \operatorname{erf}(2y)$, which has the correct wings, except case C, which uses the deliberately poor constant prior $a^* = 0.15$:

- (A) complete, exact data;
- (B) incomplete data, only 20 strikes observed;
- (C) complete, exact data with the poor constant prior;
- (D) complete data with 0.1% noise;
- (BD) incomplete and noisy.

The exact-data cases use a single small $\beta = 10^{-6}$. The noisy cases (D, BD) select β with the decreasing- β schedule and Morozov discrepancy principle described above, starting from $\beta_0 = 10^2\delta$ and warm-starting each minimization from the previous one.

Figure 6 shows the recovered coefficient for the five cases. With complete exact data (A) the reconstruction tracks both the oscillation and the sharp step over the informative range $y \in [-1.3, 1.3]$ and returns to 0.20 in the wings. Incomplete data (B) is almost as good near the observed strikes. Case C isolates the effect of the prior: the fit is still accurate where the data are informative, but in the wings it settles on the wrong prior value 0.15 rather than the true 0.10/0.20, since prices there carry almost no information about the volatility. The noisy cases (D, BD) are less accurate, and BD, which is both incomplete and noisy, is the hardest and the most sensitive to the prior, matching the behaviour reported in [6].

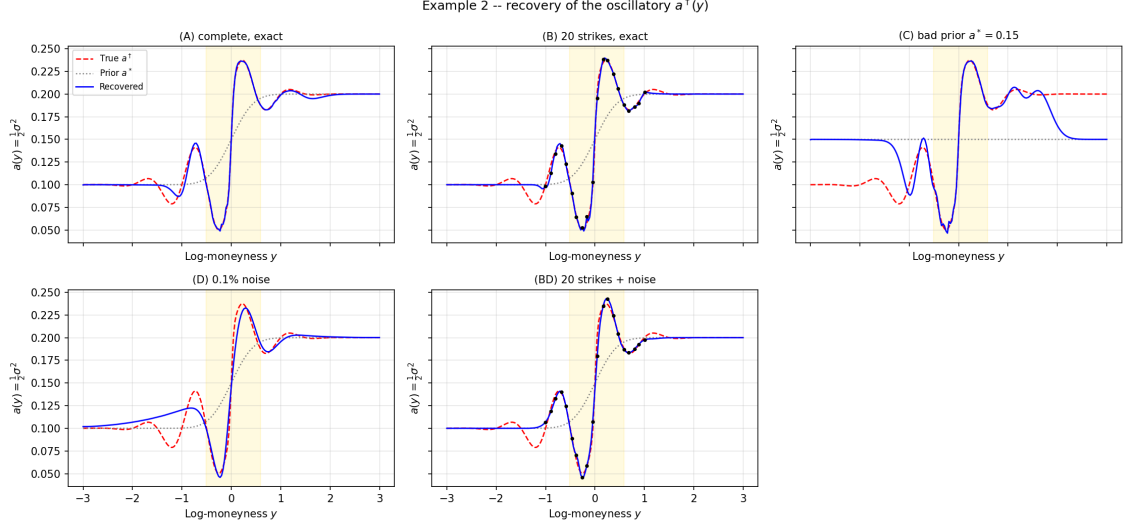


Figure 6: Example 2 of [6]: recovery of the oscillatory local variance $a^\dagger(y)$ (52) (red dashed) for the five test cases. Blue: recovered $a(y)$; grey dotted: prior a^* ; black dots (B, BD): observed strikes; shaded band: the strikes $K = 600$ – 1800 tabulated in the paper.

5.4.11 Alternative approach: Interpolation/Extrapolation of the IV surface

The Tikhonov approach above inverts the forward pricing operator directly and is robust to the noise and sparsity of market quotes, at the cost of solving a regularized optimization problem. A computationally lighter alternative is to first construct a smooth, arbitrage-free *implied volatility surface* from the quoted options-by interpolating in the liquid region and extrapolating into the wings-convert it to a continuous call-price surface $C(K, T)$ through the Black-Scholes formula, and then read off the local volatility by applying the Dupire formula (37) pointwise. Here the regularization is implicit, supplied by the smoothness of the chosen interpolant rather than by an explicit penalty, and the no-arbitrage conditions of the previous section must be enforced on the fitted surface for the resulting local variance to stay positive.

We build such a surface from market data and use it as the input to the Dupire formula; the full construction-risk-free curve, dividend-yield calibration, and the out-of-the-money composite from which the smiles are assembled-is detailed in Appendix A.

A Implied Volatility Surface - Implementation Details

This appendix describes the pipeline used to construct the SPY implied volatility surface shown in Section 3.2, including the practical difficulties encountered and the assumptions made at each step. All code lives in Implied Volatility Code and is reproducible from the cached market snapshot dated 2026-04-30.

A.1 Data

SPY option chains are fetched via the `yfinance` library. We restrict to maturities $4 \text{ wk} \leq T \leq 1 \text{ yr}$ and require a minimum trading volume of 10 contracts per quote. When both bid and ask are positive the mid-price is used; otherwise the last traded price is taken as a fallback (relevant for pre-market snapshots where the spread is stale but the last price is more informative). Data for calls and puts are cached separately as *parquet* files so that all subsequent computations are fully reproducible without network access.

A.2 Risk-free rate

The risk-free rate is taken from the US Treasury par yield curve published daily by the US Department of the Treasury. Par yields are available at standard maturities (1M, 2M, 3M, 4M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y) and are converted to continuously-compounded rates via

$$r_{cc} = \ln\left(1 + \frac{y}{100}\right),$$

where y is the par yield in percent. Rates are linearly interpolated (in T) between the published nodes and held flat outside the range. Figure 7 shows the resulting curve on the reference date.

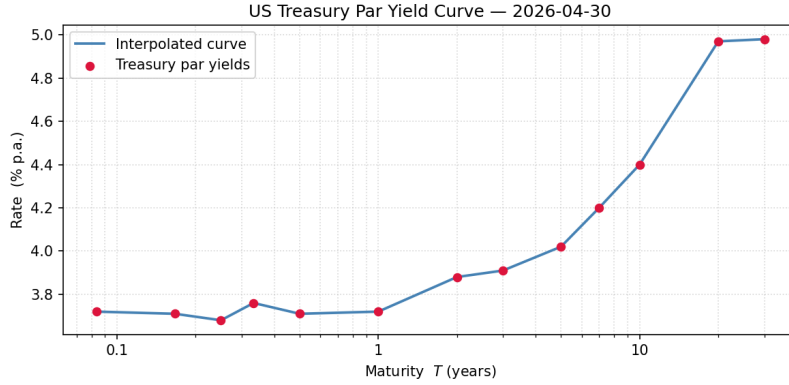


Figure 7: US Treasury par yield curve on 2026-04-30 (log-maturity axis). Red dots are published nodes; the line is the linear interpolant used to assign a per-maturity discount rate.

Note that the risk-free rate could be refined further as the US Treasuries are not strictly risk-free, as they carry a small credit risk premium. A better discounting rate for SPY options would be the overnight index swap (OIS) rate rather than the Treasury curve, but for our purposes the difference is not enough to justify the additional complexity of fetching and processing OIS data.

A.3 Dividend yield calibration

SPY pays dividends quarterly, so any pricing formula must account for the dividend yield q . Getting q wrong breaks the internal consistency of the surface: puts and calls at the same strike and maturity imply different volatilities, producing a visible discontinuity at the money in any composite surface. We went through three successive approximations before reaching a satisfactory result.

Stage 1 - $q = 0$. The simplest assumption is to ignore dividends entirely. This is equivalent to treating SPY as a non-dividend-paying asset, which is a standard simplification when the dividend yield is small relative to the interest rate and the maturity is short. For maturities beyond a few months, however, cumulative dividends are non-negligible (SPY's trailing yield is approximately 1.1%) and the forward price implied by put-call parity,

$$F = Se^{(r-q)T},$$

is overestimated if q is set to zero. As a result, the Black-Scholes-Merton formula assigns different implied volatilities to calls and puts at the same strike, since each type is sensitive to the forward in a different direction.

Stage 2 - Fixed annual yield. The natural correction is to use the trailing twelve-month dividend yield reported by the data provider ($q \approx 1.14\%$ for SPY on the reference date). This constant is applied uniformly to all maturities. The surface improves near the money, but the fix is imprecise: SPY dividends are paid on a quarterly schedule, so the effective yield between two option expirations depends on how many ex-dividend dates fall within that window. A short-dated option expiring before the next ex-dividend date carries an effective $q \approx 0$, while a longer-dated option spanning several dividend payments accumulates a yield closer to the annual figure. Applying a single annual constant systematically over-corrects for short maturities and under-corrects in an uncontrolled way across the term structure.

Stage 3 - Put-Call parity calibration. The cleanest approach is to infer q at each expiration directly from market prices using put-call parity, keeping the Treasury rate r fixed. For each matched call-put pair at strike K_i and expiration T , put-call parity gives

$$C_i - P_i = Se^{-qT} - K_i e^{-rT}.$$

Defining the cost-of-carry forward $F = Se^{(r-q)T}$ and multiplying both sides by e^{rT} yields an implied forward per pair,

$$F_i = (C_i - P_i) e^{rT} + K_i,$$

where q is encoded inside F_i rather than appearing explicitly. Taking the median over the relevant pairs at a given expiration yields a robust forward estimate $\tilde{F}$, and inverting $\tilde{F} = Se^{(r-q)T}$ gives

$$q = r - \frac{\ln(\tilde{F}/S)}{T}.$$

A key implementation choice is *which pairs to include*. Put-call parity holds model-free for any European options, but the quality of the forward estimate depends on how cleanly each leg is priced. For a strike $K < S$ the call is in-the-money and the put is out-of-the-money; for $K > S$ the roles are reversed. Deep in-the-money options are significantly

less liquid: their price is dominated by intrinsic value, the bid-ask spread is wide relative to the time value, and last-trade prices can be hours stale. Feeding these pairs into the forward estimate biases $\tilde{F}$ toward noisy values.

We therefore restrict the calibration to near-ATM pairs satisfying $|K/S - 1| \leq 0.05$ (i.e. strikes within $\pm 5\%$ of the current spot). In this band, both the call and the put are close to at-the-money; both legs have tight spreads and meaningful open interest, and the time value is large relative to the bid-ask noise. The median over these pairs (rather than the mean or a least-squares fit) guards against the occasional stale quote even in the near-ATM region. If fewer than two near-ATM pairs are available the constraint is relaxed to all matched pairs.

Since r is fixed from the Treasury curve, there is only one free parameter per expiration, which avoids the instability of simultaneously fitting both r and q . Calibrated values are bounded to $q \in [-0.02, 0.10]$; if the result is outside this range (a sign of too few matched pairs or erroneous data), the provider value is used as a fallback.

The resulting term structure shows $q \approx 0$ for near-term expirations (the next quarterly dividend has not yet been declared within the window) and a gradual convergence to the trailing annual yield for longer maturities, which is economically consistent with the quarterly payment schedule.

Figure 8 illustrates the three stages for the December 2026 expiration ($T \approx 0.63$ yr), where the effect of the dividend yield is large enough to be clearly visible. Each panel overlays the put-implied and call-implied volatility curves across the full strike range. With $q = 0$ (Stage 1) the two curves diverge significantly and the level is shifted upward for OTM puts and downward for OTM calls. With a fixed annual q (Stage 2) the ATM region improves but the term-structure mismatch remains. With the PCP-calibrated q (Stage 3) the two curves sit on top of each other near the money, confirming that the composite surface can be joined smoothly. The residual divergence at deep in-the-money strikes visible in all three panels is due to liquidity issues (wide bid-ask spreads, stale prices) rather than the dividend yield assumption, and is removed by restricting the composite to out-of-the-money quotes only.

A.4 OTM composite construction

Once r and q are consistent with market prices, the composite surface is formed by taking out-of-the-money options only: OTM puts ($K < S$) for the left wing and OTM calls ($K \geq S$) for the right wing.

The rationale is that in-the-money options are significantly less liquid than their out-of-the-money counterparts: ITM bids tend to be stale, spreads are wide, and the option price is dominated by intrinsic value, so a small absolute pricing error translates into a large relative error in the time value — and therefore a large error in the inverted implied volatility. This is visible in Figure 8: the divergence between call and put IV curves away from the money persists even in Stage 3, not because q is wrong but because deep-ITM prices are unreliable.

Figures 9 and 10 show the full call and put surfaces (all strikes, no ITM filter) to illustrate the noise. The clean composite (Figure 3) is obtained after restricting to $0.75 \leq K/S \leq 1.25$ and keeping only the OTM leg for each option type.

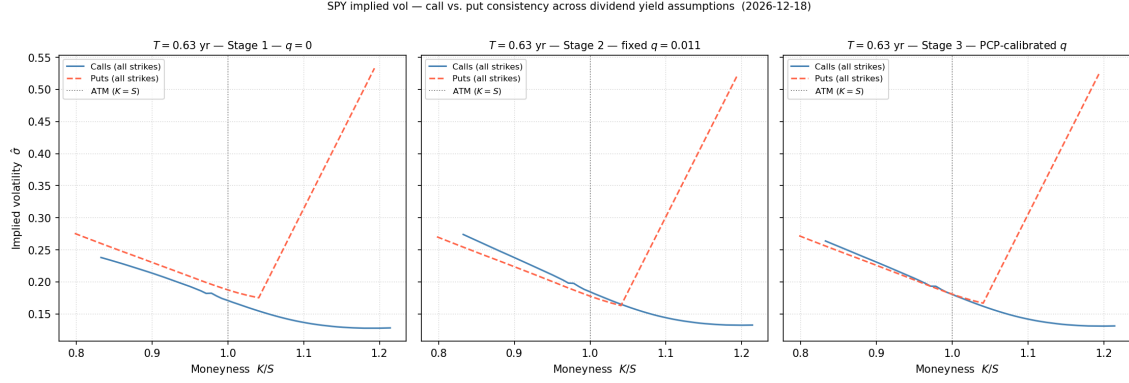


Figure 8: Call-implied (solid blue) and put-implied (dashed red) volatility across all available strikes at the December 2026 expiration ($T \approx 0.63$ yr), under three dividend yield assumptions. Stage 1 ($q = 0$): curves diverge sharply — the forward is overestimated, shifting call and put IVs in opposite directions. Stage 2 (fixed annual q): improved near ATM, but the flat term structure leaves systematic errors for maturities where the true q differs from the annual average. Stage 3 (PCP-calibrated q from near-ATM pairs): the two curves coincide in the OTM region, where both legs are liquid. The residual divergence far from ATM (deep ITM options) is a liquidity artefact — those strikes are excluded from the composite surface.

A.5 Expiration selection

SPY has a dense cluster of weekly expirations in the short end (roughly 1-8 weeks). Including all available expirations would over-weight this region and produce a surface that appears smooth in absolute maturity but poorly samples the longer end. To obtain a balanced representation, the full maturity range is divided into $N = 10$ log-spaced bins; within each bin the expiration with the highest combined open interest (calls plus puts) is kept. This ensures that liquidity guides the selection and that the resulting sample is approximately uniform in $\log T$, which is the natural scale on which the term structure of volatility varies.

A.6 Summary of assumptions

For reference, the key modelling assumptions are:

- **European exercise:** SPY options are technically American-style. The early-exercise premium is small for an index ETF with a low continuous yield and is neglected here; the BSM formula is used as an approximation.
- **Continuous dividends:** actual SPY dividends are discrete quarterly payments. The PCP calibration implicitly converts them into a continuous yield that is consistent with the current option prices, which is the standard market convention.
- **Moneyness filter** $0.75 \leq K/S \leq 1.25$: strikes outside this range are too illiquid to contribute reliable IV observations.
- **Volume filter:** quotes with fewer than 10 contracts of daily volume are excluded.
- **Single snapshot:** the surface is computed from a single end-of-day snapshot; intraday bid-ask movements are not captured.

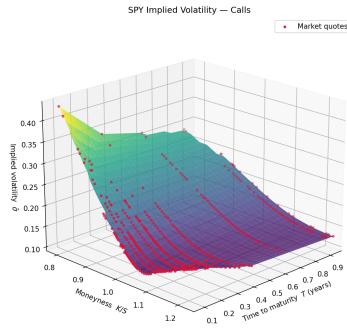


Figure 9: IV surface from all call quotes (including deep ITM, $K \ll S$). Deep ITM calls carry noisy implied volatilities due to wide spreads and stale prices.

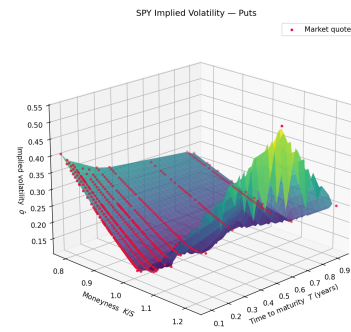


Figure 10: IV surface from all put quotes (including deep ITM, $K \gg S$). Same liquidity issue for strikes far above the spot.

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